

The Binary Black Hole Merger Rate Deviates From the Cosmic Star Formation Rate: A Tug of War Between Metallicity and Delay Times

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ABSTRACT

Gravitational-wave detectors are now making it possible to investigate how the merger rate of binary black holes (BBHs) evolves with redshift. In this study, we examine whether the BBH merger rate of isolated binaries deviates from a scaled star formation rate density (SFRD)—a frequently used model in state-of-the-art research. To address this question, we conduct population synthesis simulations using COMPAS with a grid of stellar evolution models, calculate their cosmological merger rates, and compare them to a scaled SFRD. We find that our simulated rates deviate by factors up to $3.5\times$ at $z \sim 0$ and $5\times$ at $z \sim 9$ due to two main phenomena: (i) The formation efficiency of BBHs is an order of magnitude higher at low metallicities than at solar metallicity; and (ii) BBHs experience a wide range of delays (from a few Myr to many Gyr) between formation and merger. Deviations are similar when comparing to a *delayed* SFRD, and even larger (up to $\sim 10\times$) when comparing to SFRD-based models scaled to the local merger rate. Interestingly, our simulations find that the BBH delay time distribution is redshift-dependent, increasing the complexity of the redshift distribution of mergers. We find similar results for simulated merger rates of BHNSs and BNSs. We conclude that the rate of BBH, BHNS, and BNS mergers from the isolated channel can significantly deviate from a scaled SFRD, and that future measurements of the merger rate will provide insights into the formation pathways of gravitational-wave sources.

Keywords: Gravitational waves (678) — Binary stars (154) — Compact objects (288)

1. INTRODUCTION

The rapidly increasing sample of gravitational wave (GW) events detected by the Advanced LIGO and Virgo interferometers offers a new opportunity to explore the formation and properties of black holes (BHs) and neutron stars (NSs) as a function of redshift. The most recent GW catalogs (GWTC-3, OGC-4) and independent GW data analyses already contain about 100 binary black hole (BBH) mergers out to redshifts $z \sim 1.5$ (Abbott et al. 2019, 2021, 2023, 2024; Nitz et al. 2023; Venumadhav et al. 2019, 2020; Zackay et al. 2019; Olsen et al. 2022; Mehta et al. 2023; Wadekar et al. 2023), and next-generation GW detectors, such as Cosmic Explorer and Einstein Telescope, are poised to detect stellar-mass black hole mergers beyond $z \gtrsim 10$ (e.g., Punturo et al. 2010; Sathyaprakash et al. 2012; Reitze et al. 2019; Maggiore et al. 2020; Evans et al. 2021, 2023; Singh et al.

2022; Gupta et al. 2023). The rate and properties of BBH, black hole–neutron star (BHNS), and binary neutron star (BNS) mergers as a function of redshift can provide invaluable insights into the physical processes underlying BH and NS formation, the massive (binary) stars that lead to their formation, their host galaxies, and the different formation channels at play (e.g., Fishbach et al. 2018; Rodriguez & Loeb 2018; Vitale et al. 2019; Bavera et al. 2021; Ng et al. 2021; Chu et al. 2022; Chruślińska 2022; Mapelli et al. 2022; van Son et al. 2022; Ray et al. 2023; Santoliquido et al. 2023; Vijaykumar et al. 2023). A challenge, however, is that information about the formation pathway and progenitor system is not directly imprinted in the GW observations. Inferring such properties, or making predictions for future GW detectors, therefore often requires making assumptions about the underlying merger population.

A common assumption in the literature is that the BBH (and BHNS/ BNS) merger rate, $\mathcal{R}_{\text{merge}}(z)$, can be described by scaled versions of the cosmic star formation rate density (SFRD), matched to the local observed merger rate (e.g., Iacovelli et al. 2022; Gupta et al. 2023; Lehoucq et al. 2023). This assumption is based on the notion that GW sources are formed from massive stars, whose formation rate is described by the SFRD, which rapidly form binary compact object systems (~ 10 Myr; e.g., Kippenhahn & Weigert 1990) that merge after some time. There are many astrophysical processes in the evolution of binary stars, however, that can drastically alter this paradigm, leading to a merger rate that might *not* follow a scaled SFRD. As will be the focal point of this paper, there are two key processes within the isolated binary evolution formation channel that can cause significant deviations from the SFRD.

First, the formation efficiency of compact object binaries is metallicity-dependent. The evolutionary outcome of massive (binary) stars can rely strongly on birth metallicity, as metallicity drives mass loss through stellar winds which impacts, for example, radial extension of stars and the remnant mass of compact objects (e.g., Vink et al. 2000, 2001; Langer 2012). Recent work has shown that the formation yield of BBHs is strongly dependent on their birth metallicity (Chruslinska et al. 2018; Giacobbo & Mapelli 2018; Broekgaarden et al. 2022), resulting in a BBH merger rate that does not follow the SFRD, but an SFRD convolved with this metallicity dependence.

Second, BBH mergers may occur with a significant delay relative to the formation epoch, and the distribution of delay times can itself be metallicity-dependent (e.g., Belczynski et al. 2016a; Giacobbo & Mapelli 2018). Although massive binary stars form BBH systems (or BHNS and BNS systems) in just ~ 10 Myr, the merger delay times can be as long as many Gyr (e.g. Peters 1964; Neijssel et al. 2019), resulting in a merger rate that may follow a *delayed* SFRD. Indeed, some GW studies model $\mathcal{R}_{\text{merge}}(z)$ as a scaled SFRD combined with a delay time distribution, which is often assumed to follow a simple power law, $dn/dt \propto t^{-1}$, with a minimum delay time of $\sim 10 - 100$ Myr (e.g., Regimbau et al. 2012; Belgacem et al. 2019; Borhanian & Sathyaprakash 2022; Colombo et al. 2022; Iacovelli et al. 2022; Naidu et al. 2022; Lehoucq et al. 2023).

Here, we investigate how the two aforementioned effects impact $\mathcal{R}_{\text{merge}}(z)$. We employ a large set of population synthesis models for the isolated binary evolution formation pathway to simulate BBH (as well as BHNS and BNS) mergers and investigate the resulting merger rates as a function of redshift for a wide range of as-

sumptions about stellar and binary evolution. The paper is structured as follows. We describe our models and methods in Section 2. We investigate the effects of metallicity-dependent formation yield and delay times on $\mathcal{R}_{\text{merge}}(z)$ in Section 3.1 and Section 3.2, respectively. In Section 3.3, we assess the accuracy of models from the literature by comparing our simulated rates to SFRD-based models scaled to the inferred local BBH merger rate and convolved with different delay time distributions. We discuss our results for BHNS and BNS mergers in Section 3.4. We end with a discussion in Section 4 and a summary of our key findings in Section 5.

2. METHODS

2.1. Population Synthesis Simulated Merger Rates

We use the simulation output and methodology from Boesky et al. (2024), which are briefly summarized here. Boesky et al. (2024) presents a population of simulated compact object mergers formed through the isolated binary evolution pathway. The simulations are generated using COMPAS¹ (Team COMPAS: Riley et al. 2022), a rapid population synthesis code. COMPAS simulates the evolution of massive stars based on analytical fitting formulae for single and binary star evolution by Hurley et al. (2000, 2002), derived from stellar evolution tracks presented in Pols et al. (1998), as well as earlier work by Eggleton et al. (1989) and Tout et al. (1996).

Boesky et al. (2024) studies two two-parameter-varied grids of simulations. Here, we only focus on the grid that varies the common-envelope (CE) efficiency, α_{CE} , and mass transfer efficiency, β , parameters. The CE phase results from dynamically unstable mass transfer in which the companion star is engulfed in the donor’s envelope and tightens the binary through drag (see Ivanova et al. 2020, and references therein). COMPAS employs the ‘ $\alpha_{\text{CE}} - \lambda$ ’ formalism (Webbink 1984; de Kool 1990) to parameterize the CE phase. The α_{CE} parameter is of particular interest when investigating the merger rate because it is capable of significantly reducing binary separation, and thus potentially altering the expected $dn/dt \propto t^{-1}$ delay time distribution (e.g. Belczynski et al. 2018; Vigna-Gómez et al. 2018). Our grid uses four CE efficiency values, $\alpha_{\text{CE}} = 0.1, 0.5, 2$, and 10 , which is a range representative of prior studies (e.g., Vigna-Gómez et al. 2018; Bavera et al. 2021; Dorozsmai & Toonen 2024; Santoliquido et al. 2021; Broekgaarden et al. 2022). In tandem with α_{CE} , we also vary the accretion efficiency $\beta = \Delta M_{\text{acc}}/M_{\text{donor}}$, where ΔM_{donor}

¹ Compact Object Mergers: Population Astrophysics and Statistics, <https://compas.science>

and ΔM_{acc} are the changes in the mass of the donor and accretor stars, respectively. We use $\beta = 0.25, 0.5$, and 0.75 to reflect the parameter’s theoretical range of $[0, 1]$ cf. earlier work (Broekgaarden et al. 2022; van Son et al. 2022; Dorozzmai & Toonen 2024). Henceforth, we refer to the model with $\alpha_{\text{CE}} = 2$ and $\beta = 0.5$ as the “fiducial” model.

Throughout our study, we adopt a popular model of the SFRD from Madau & Dickinson (2014) (Equation 15):

$$\psi(z) = a \frac{(1+z)^b}{1 + [(1+z)/c]^d} \text{ M}_\odot \text{ yr}^{-1} \text{ Mpc}^{-3}, \quad (1)$$

with $a = 0.01$, $b = 2.6$, $c = 3.2$, and $d = 6.2$ (Madau & Fragos 2017). The true SFRD is uncertain, and exploring the impact of assumed SFRD model is left for future research. We expect the qualitative results in this work, however, to be robust under different choices for the SFRD.

To calculate the merger rate of a simulated binary population of BBH, BHNS, or BNS mergers, we convolve a metallicity-specific SFRD, $\mathcal{S}(Z_i, z)$, with the formation rate of the population:

$$\begin{aligned} \mathcal{R}_{\text{merge}}(t_m, M_1, M_2) &\equiv \frac{d^4 N_{\text{merge}}}{dt_m dV_c dM_1 dM_2}(t_m, M_1, M_2) \\ &= \int dZ_i \int_0^{t_m} dt_{\text{delay}} \mathcal{S}(Z_i, z(t_{\text{form}} = t_m - t_{\text{delay}})) \times \\ &\quad \frac{d^4 N_{\text{form}}}{dM_{\text{SFR}} dt_{\text{delay}} dM_1 dM_2}(Z_i, t_{\text{delay}}, M_1, M_2), \end{aligned} \quad (2)$$

where t_m is the time of the merger in the comoving frame, t_{delay} is the time between the formation and merger of a binary, and M_1 and M_2 are the binary component masses. To obtain the metallicity-dependent star formation rate $\mathcal{S}(Z_i, z(t_{\text{form}}))$, we convolve the commonly-used galaxy stellar mass function from Panter et al. (2004) with the mass-metallicity relation from Ma et al. (2016), and multiply the result with the SFRD in Equation 1 (cf. Neijssel et al. 2019; Broekgaarden et al. 2022). We use `astropy` to transform t_m to a redshift using the WMAP9 cosmology². For more details on the merger rate and its calculation, see the methodology in Team COMPAS: Riley et al. (2022).

2.2. SFRD-Based Toy Model Merger Rates

² We find that the rates are not significantly impacted by the choice of cosmology (cf. Neijssel et al. 2019).

To investigate whether popular, simplistic models of the merger rate are representative of population synthesis results, we compare our simulations to three models from the literature. Throughout this study, we will refer to these as “toy models”. The first, and simplest, toy model is a scaled SFRD. The second is a scaled SFRD convolved with a constant delay time of $t_{\text{delay}} = 20$ Myr. The third is a SFRD convolved with a $dn/dt \propto 1/t$ delay time distribution that has a minimum delay time of ~ 20 Myr (similar to studies including Regimbau et al. 2012; Borhanian & Sathyaprakash 2022; Colombo et al. 2022; Iacovelli et al. 2022; Lehoucq et al. 2023).

For all toy models, we use the SFRD described in Equation 1 to match the SFRD assumed for calculating the merger rate. Throughout this paper, we compare toy models to our population synthesis simulated merger rates (Equation 2) by normalizing both such that their areas integrate to unity, and then dividing the simulated $\mathcal{R}_{\text{merge}}(z)$ by the toy model. We first discuss results for the scaled SFRD toy model, and then we describe results for the two delayed SFRD toy models in Section 3.3 and beyond.

3. RESULTS: DEVIATION FROM THE SFRD

In Figure 1, we show the BBH merger rates produced with our suite of models, and compare them to an arbitrarily scaled SFRD. In the left panel, we find that our simulated models produce different merger rate distributions as a function of redshift, and also lead to different merger rate magnitudes. Models with $\alpha_{\text{CE}} = 0.5, 2.0$ produce merger rates of factors $\sim 3 - 10$ higher than models with $\alpha_{\text{CE}} = 0.1, 10.0$. This is a result of $\alpha_{\text{CE}} = 0.1$ causing binaries to merge before stars are able to become BBHs, while $\alpha_{\text{CE}} = 10$ does not tighten the binaries enough to merge within a Hubble time³.

Most importantly for the scope of this study, we find that the redshift evolution of our simulated merger rates deviate from the scaled SFRD toy model. These deviations from the SFRD are highlighted in the right panel of Figure 1, in which we normalize the merger rates and divide them by the normalized SFRD⁴. We find that the normalized BBH merger rates tend to be higher (up to a factor of 3.5) compared to the SFRD at low redshift $z \lesssim 1.5$, as is visible by all lines exceeding unity. The merger rates approach the scaled SFRD around redshift 2, and then fall below the SFRD above redshifts $\gtrsim 2$,

³ We elaborate on this “sweet-spot” of α_{CE} in Section 3.2.

⁴ We note that this normalization is chosen to investigate the *relative* differences between our simulations and the SFRD toy model, but leads in practice to minimal differences around the merger rate peak $z \sim 2$. See Section 3.3 for results that scale by the local merger rate instead.

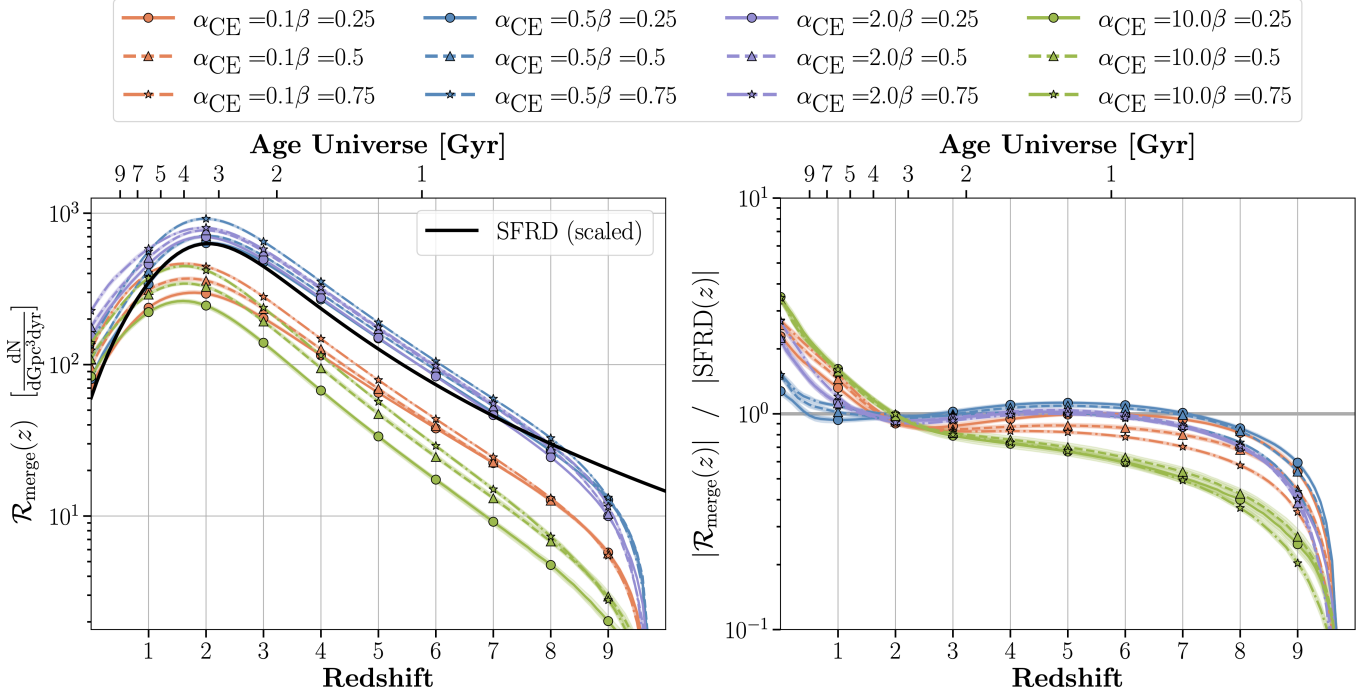


Figure 1. *Left:* The BBH merger rate as a function of redshift for the α_{CE} and β model variations explored in this paper compared to the SFRD scaled arbitrarily (black line). *Right:* The BBH merger rate divided by the SFRD, where we normalize the areas of both to 1. The horizontal grey line indicates a merger rate that follows the SFRD. In both panels we include the 1σ and 2σ confidence intervals calculated by bootstrapping the simulation results to show the sampling uncertainty (note that uncertainties are typically the same size as the width of the lines). The sharp decline in the merger rate at $z \sim 9$ is due to star formation starting at $z = 10$ in our calculations and the fact that mergers are delayed.

with the exact z at which this occurs depending on the model. Some models only fall below the SFRD line at high redshifts $z \gtrsim 7$, such as those with $\alpha_{\text{CE}} = 0.5$. We find deviations of our simulated merger rates from the SFRD as high as factors $5\times$ (for the $\alpha_{\text{CE}} = 10$ models around $z \sim 9$). The first three columns in Table 1 in the appendix provide statistics on the relative ratios between the (normalized) simulated BBH merger rates and SFRD from $z = 0$ to $z = 9$.

Despite model-to-model variations, the general trend among our simulated merger rates in Figure 1 indicates shared underlying physical processes that cause $\mathcal{R}_{\text{merge}}(z)$ to deviate from the SFRD. As stated in Section 1, the physical sources of merger rate-SFRD deviations is the effect of metallicity and delay times, which we explore comprehensively in the following two sections.

3.1. The Effect of Metallicity on the Merger Rate

First, we focus on the effect of the metallicity-dependent formation efficiency on $\mathcal{R}_{\text{merge}}(z)$ for BBHs. In the left panel of Figure 2, we plot the BBH formation rate as a function of redshift, $\mathcal{R}_{\text{form}}(z)$ (Equation 1 in Broekgaarden et al. 2021), for all our models and in the right panel of Figure 2, we plot the ratio of

BBH formation rate relative to the scaled SFRD (the toy model). If the formation yield of BBHs were independent of metallicity, the formation rate of BBHs as a function of redshift would exactly follow the SFRD since the simulations assume all other initial conditions (e.g., initial mass, initial separation) to be independent of initial metallicity, and the time required for massive stars to form BBHs is negligible on this scale. We find the following results.

First, we observe that, similarly to $\mathcal{R}_{\text{merge}}(z)$ in Figure 1, models with $\alpha_{\text{CE}} = 0.5, 2.0$ tend to have higher $\mathcal{R}_{\text{form}}(z)$ than the other two α_{CE} values by a factor of up to ~ 5 . Notably, there is very little variation in the redshift evolution of $\mathcal{R}_{\text{form}}(z)$ between models, and they all deviate from the scaled SFRD. Specifically, the peak of the BBH formation rate from all models is at a higher redshift, $z \approx 2.5$, than the peak of the SFRD, $z \approx 2$. As a result, we find that the BBH formation rate in the simulated $\mathcal{R}_{\text{form}}(z)$ is suppressed at $z \lesssim z_{\text{peak}}$ compared to the scaled SFRD with deviations of up to an order of magnitude at $z \sim 0$. The simulated BBH rate is relatively boosted for $z \gtrsim z_{\text{peak}}$ by a factor of $\sim 2\times$ compared to the SFRD. All in all, our results find a metallicity-dependence in the formation rates of population synthesis simulations which favors high z BBH

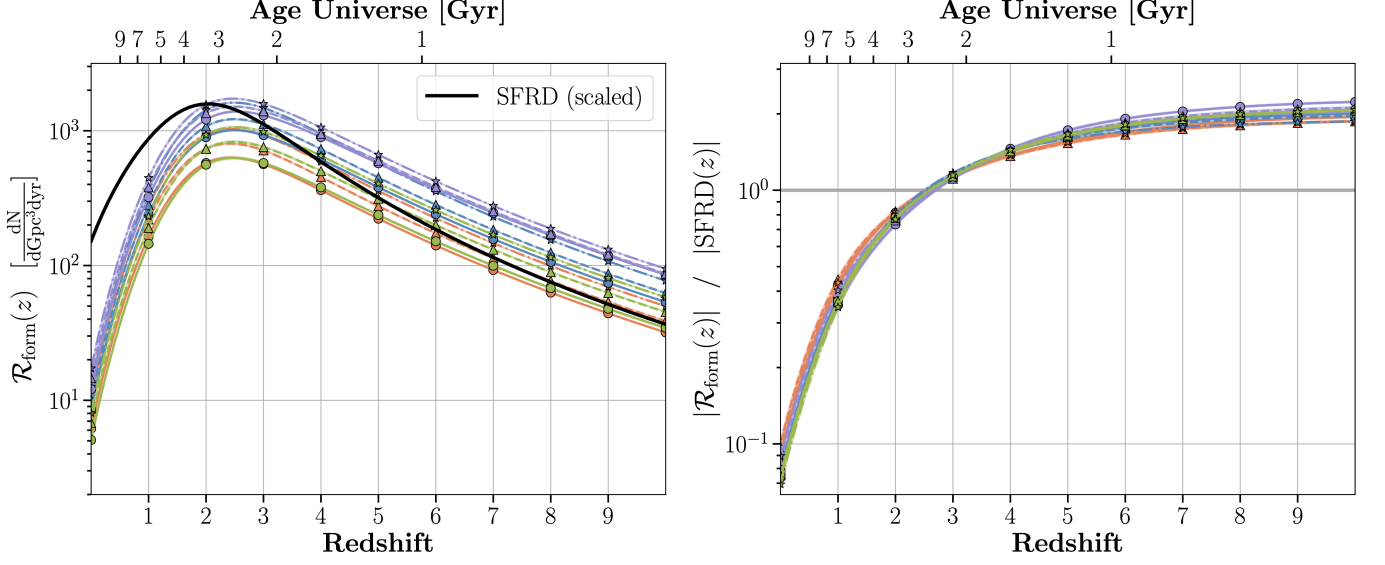


Figure 2. Same as Figure 1, but for the BBH *formation* rate (formation is defined as the moment when the second BH forms). The peak of the BBH formation rate is clearly shifted to higher redshifts than the SFRD due to a higher BBH formation efficiency at lower metallicities.

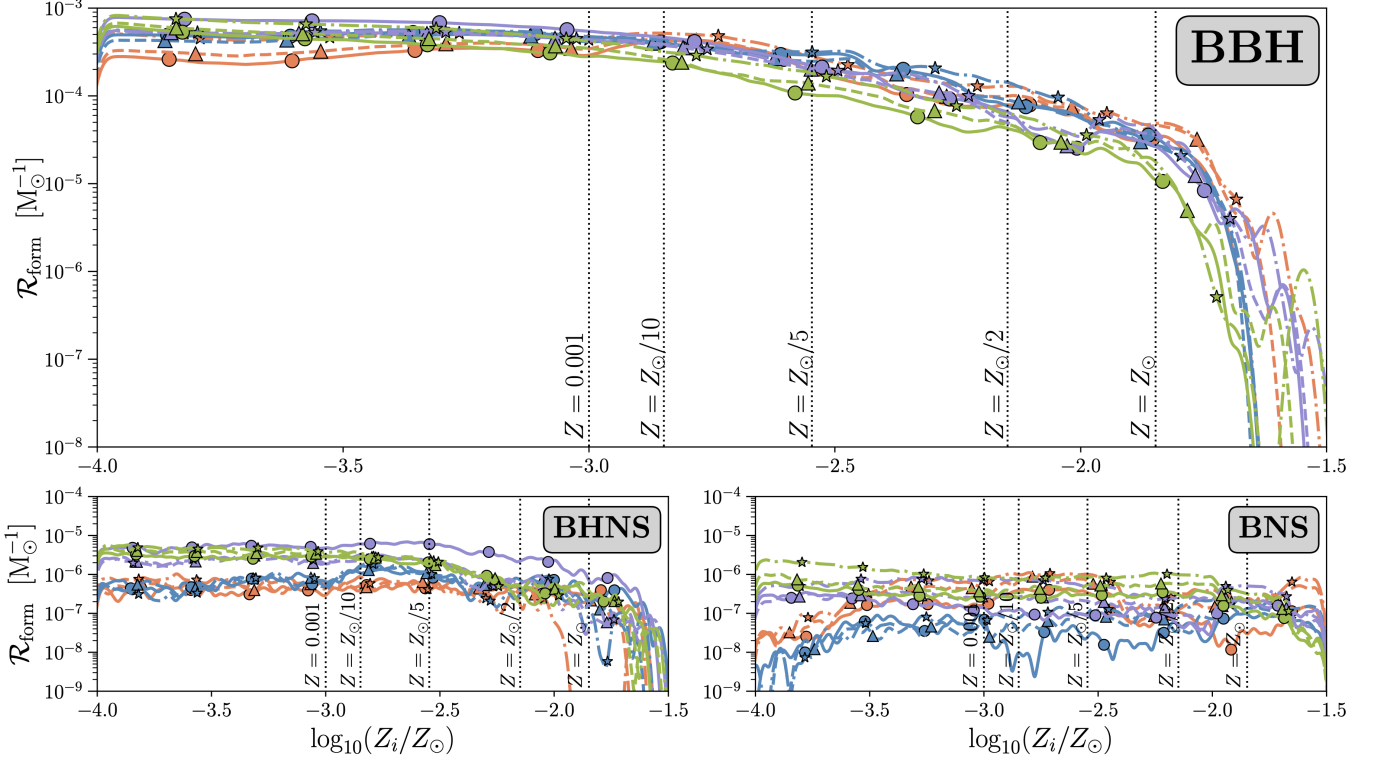


Figure 3. The formation yield of BBHs, BHNSs, and BNSs that will merge within a Hubble time, per solar mass of star formation, $d\mathcal{R}_{\text{form}}/dM_{\text{SFRD}}$, as a function of birth metallicity, Z_i , for all models. We choose initial metallicities for each binary by sampling randomly from a log-uniform metallicity distribution from $Z_i = 0.0001$ to $Z_i = 0.03$. We then obtain $\mathcal{R}_{\text{form}}(Z_i)$ using a kernel density estimation over the initial metallicities of the simulated binaries. The decline in BBH formation efficiency at $Z_i \gtrsim Z_{\odot}/5$ is clearly visible.

formation, thus favoring the rate of *merging* BBHs at higher redshifts compared to lower redshifts.

To further probe the metallicity dependence of BBH formation, we show the simulated formation rates of BBHs as a function of initial metallicity, Z_i , in the top panel of Figure 3. We find that as Z_i increases, the BBH formation yield falls off more and more rapidly, as much as an order of magnitude by $Z_\odot/2$, and three orders of magnitude by $2Z_\odot$. This rapid decline is consistent with findings for BBH mergers formed from isolated binary evolution in recent literature (e.g., [Giacobbo & Mapelli 2018](#); [Klencki et al. 2018](#); [Giacobbo et al. 2018](#); [Chruslinska et al. 2018](#); [Neijssel et al. 2019](#); [Santoliquido et al. 2021](#); [Broekgaarden et al. 2022](#), and [Dorozmai & Toonen 2022](#)). Since the mean metallicity increases with cosmic time, higher BBH formation efficiency at low progenitor metallicity results in a higher BBH formation rate relative to the SFRD at higher redshifts ($z \gtrsim 2.5$) (as is visible in Figure 2). The redshift at which we observe notable increases in formation rate is around where the proportion of formed stars has a considerable change in birth metallicity based on the Madau-Dickinson prescription (see Appendix B for details on the metallicity-dependent SFRD). Low model-to-model variation in the relationship between metallicity and formation yield leads to the tight spread (within factor few) of BBH formation rates among models that we observe in Figure 2 (cf. [Chruslinska et al. 2018](#); [Santoliquido et al. 2021](#); [Broekgaarden et al. 2022](#)). Boosted BBH formation efficiency at low birth metallicity is predominantly caused by less wind loss through line-driven stellar winds at low metallicity, which leads to tighter binary systems (e.g., because more mass needs to be expelled during a CE event) and fewer systems that disrupt during supernova. We discuss this in more detail in Appendix A.

3.2. The Effect of Delay Times on the Merger Rate

The delay times (t_{delay}) between the formation and merger of BBHs, can range from a few Myr to greater than the Hubble time ([Mennekens & Vanbeveren 2016](#); [Belczynski et al. 2016b](#); [Lipunov et al. 2017](#); [Steven et al. 2017](#); [Eldridge et al. 2019](#)) and lead the BBH merger rate to peak at a lower redshift than the formation rate. It is generally thought that the delay time distribution roughly follows $dn/dt \propto t^{-1}$ with a minimum delay time, t_{min} , typically bound between 10 Myr – 500 Myr (e.g., [Regimbau et al. 2012](#); [Borhanian & Sathyaprakash 2022](#); [Colombo et al. 2022](#); [Iacovelli et al. 2022](#); [Naidu et al. 2022](#)).

To understand the effect of delay times, we first investigate whether BBH delay times are dependent on

formation redshift. In the left panel of Figure 4, we show the normalized distributions of delay times for binaries formed at $z = 0.2, 2, 6$ for our fiducial model. We do indeed find redshift dependence, as the distributions for $z = 0.2$ and $z = 2$ have flatter slopes than $1/t$ —specifically, the best fits are $t^{-0.60}$, $t^{-0.77}$, and $t^{-0.95}$ for $z = 0.2, 2.0$, and 6.0 , respectively, in the region $t_{\text{delay}} > 100$ Myr. We also find that the delay time distributions at all three redshifts flatten out for delay times $\lesssim 500$ Myr (compared to $t_{\text{delay}} > 500$ Myr).

In the right panel of Figure 4 we show the distributions of delay times from our fiducial model, differentiating observable and unobservable mergers, where the latter are those formed at a given redshift with delay times longer than the time between its formation and Hubble time. We find that the majority of binaries formed at $z = 0.2$ merge after $z = 0$. Furthermore, while there are around an order of magnitude more BBH mergers at $z = 2.0$ than to $z = 6.0$, the proportion of *observable* mergers at $z = 2$ is less than that from $z = 6.0$ due to differences in the range of delay times that lead to mergers we can detect.

In Figure 5, we show how the median delay time for BBHs merging in Hubble time evolves with redshift for all models. We find two main results. First, the median delay time evolves differently as a function of redshift for different models. Among our simulations there are four sets of models with similar median t_{delay} behavior throughout redshift, which can be differentiated by α_{CE} value. This suggests that the CE mass transfer phase and the CE efficiency parameter used in the simulations are important in determining a binary’s delay time (cf. [Chu et al. 2022](#)) and that the effect of α_{CE} on the delay time distribution is redshift-dependent. We find that the median delay time in Figure 5 does not increase monotonically with the α_{CE} value. This is due to the relationship between α_{CE} and the post-CE separation: smaller α_{CE} values require the binary’s orbit to shrink more for successful envelope ejection, which in turn leads to significantly shorter delay times ([Peters 1964](#)). There is, however, a “sweet-spot” for the α_{CE} value: values that are too low prevent the CE from being successfully ejected, and the system instead undergoes a stellar merger, leading to the non-monotonic behavior (cf. [Kruckow et al. 2018](#); [Bavera et al. 2021](#); [Broekgaarden et al. 2022](#)). Second, the delay time medians are not constant throughout cosmic history, as can be seen by the decrease in the median BBH delay time as a function of redshift for nearly all models in Figure 5. This is because at higher redshifts, stars form with lower metallicities so experience weaker stellar winds and less radial expansion. These conditions typically lead binary sys-

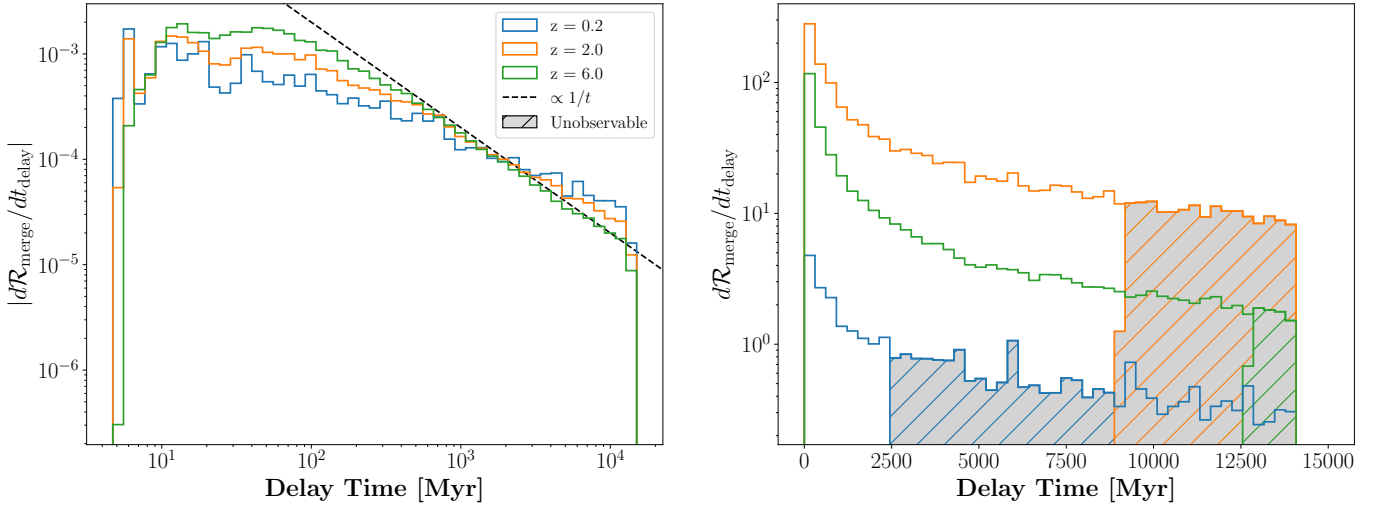


Figure 4. The BBH delay time distributions from our fiducial model for binaries formed at redshift 0.2, 2, and 6. (*Left*) The distributions normalized to an area of 1. The bin widths are log-uniformly and uniformly spaced in the left and right panels, respectively. (*Right*) The gray hatched regions are the portion of each distribution that would not merge by $z = 0$. The lowest t_{delay} bins (those with $t_{\text{delay}} \lesssim 10$ Myr) suffer from sampling uncertainty.

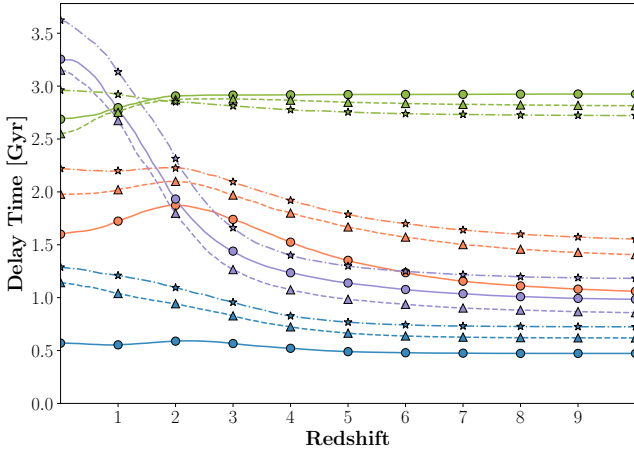


Figure 5. The median delay times of BBH mergers for each model in grid A as a function of redshift. The median only includes BBHs that merge in Hubble time for each redshift.

tems to retain more mass, experience less orbital widening, and undergo the same evolutionary pathway in a tighter orbit. Combined, this causes binaries to form BBH systems with shorter orbits at lower metallicity, which reduces the typical simulated BBH delay time at higher redshifts. The models with $\alpha_{\text{CE}} = 2.0$ have the most distinct median delay time z -evolution; this unique redshift dependence is caused by a drastic increase in the proportion of mergers formed through the classic CE channel at higher redshift that leads to shorter delay times compared to the “only stable mass transfer channel” for these simulations (in agreement with, e.g., Olejak et al. 2022; van Son et al. 2022). See Appendix D

for more details on the formation channel breakdown of our simulations throughout cosmic history.

3.3. Comparison to Delayed SFRD Models and Models Scaled to the Local Merger Rate

So far, we have presented the deviations of population synthesis simulated BBH merger rates from a scaled SFRD, but we now consider whether our simulated merger rates deviate from a *delayed* SFRD. This is motivated by—in addition to support for and use of delays in models from the literature—our findings in Section 3.2 that the delay time distribution of BBH mergers can be complex and variable between models. We use two different delayed SFRD models: (i) a scaled SFRD convolved with a delay time distribution $dn/dt \propto \delta(t - c)$, where $c = 20$ Myr, and (ii) a scaled SFRD convolved with a delay time distribution $dn/dt \propto 1/t$ with a minimum delay time of 20 Myr. See Section 2.2 for more details and references on delayed toy models.

The top panels in Figure 6 show the deviations of our simulated $\mathcal{R}_{\text{merge}}(z)$ from the SFRD model with constant t_{delay} . We find that at high redshifts ($z \gtrsim 5$), the population synthesis merger rates deviate by factors up to $\sim 3\times$ from the constant delay toy model—this is slightly less, 2% versus 4% at $z = 6$ and 1% versus 13% at $z = 7$, than deviations from the SFRD model with no delays (see Table 1). On the other hand, at low redshifts, particularly for $z \sim 0$, deviations are up to a factor $5\times$, which is larger than deviations without delay times (which has deviations up to $3\times$ at $z \sim 0$).

In the bottom panels of Figure 6 we show the deviations of our simulated BBH merger rates from a SFRD

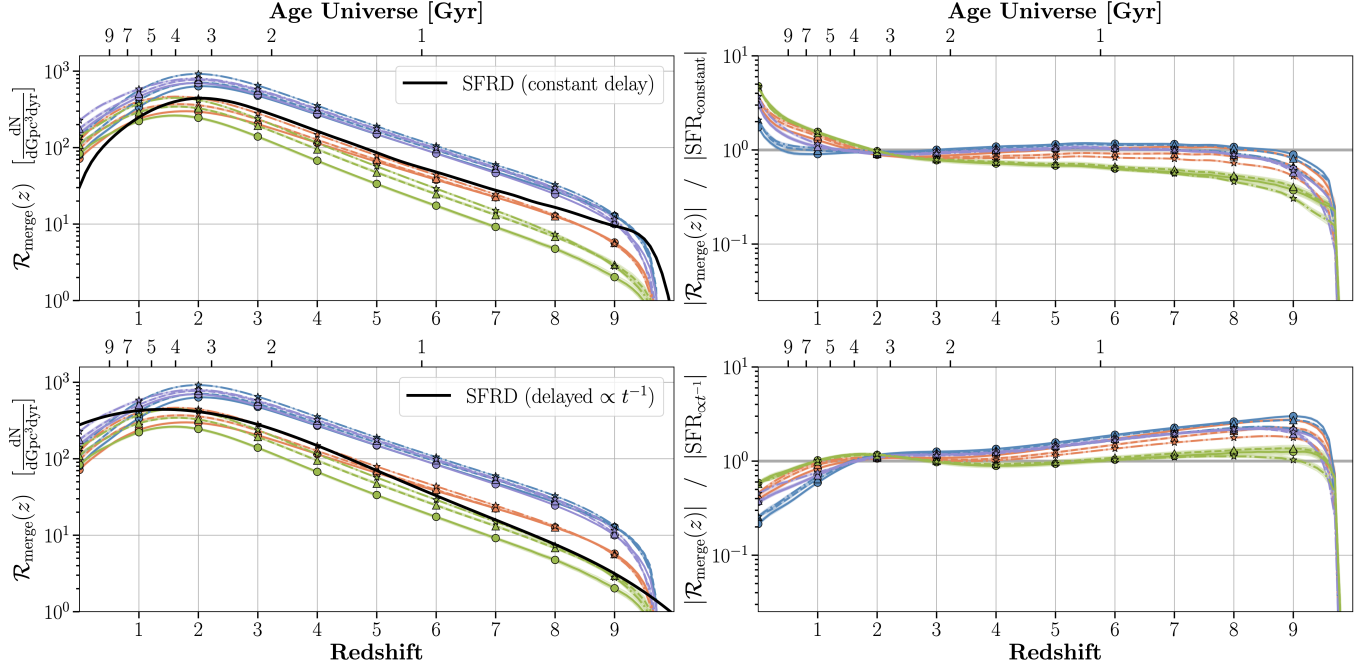


Figure 6. BBH merger rate as a function of redshift compared to toy models assuming a scaled SFRDs with a constant delay time of 20 Myr (top row) and a delay time distribution of $dn/dt \propto 1/t$ with a minimum delay time of 20 Myr (bottom row). Labels and colors are the same as in Figure 1.

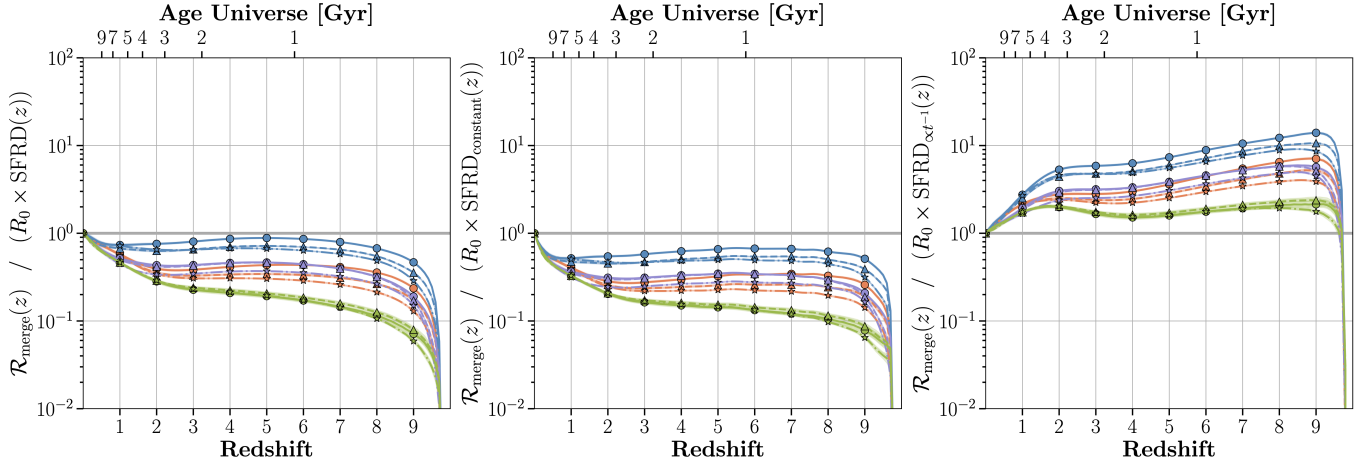


Figure 7. BBH merger rate as a function of redshift compared to toy model merger rates assuming a scaled SFRD (left), a scaled SFRD with constant delay times (middle), and a scaled SFRD with a $\propto t^{-1}$ delay time distribution (right). All simulations are normalized to the same intrinsic merger rate $R_{\text{merger},0}$. We assumed minimum delay times for the middle and right panel of 20 Myr. Labels and colors are the same as in Figure 1.

with a $1/t$ delay time distribution. This toy model peaks at a lower redshift of $z \sim 1.5$ compared to $z \sim 2$ for the non-delayed or constant delayed SFRD merger rate models. We find that our population synthesis model merger rates deviate with factors up to $5\times$ at low redshifts $z \lesssim 2$ from the $1/t$ delayed SFRD merger rate model and up to factors $3\times$ at high redshift $z \sim 9$ (see Table 1). Finally, it is important to note that there is a

considerable difference between the effect of convolving the SFRD with a $1/t$ versus constant delayed distribution; therefore, it is crucial that the high t_{delay} tail is considered by studies in the future.

Lastly, in Figure 7 we show the deviations of our simulated merger rates from the toy models where we compare by scaling to the local merger rate $R_0 \equiv R_{\text{merger}}(0)$ —a scaling often chosen for SFRD-based

merger rate models in the literature (e.g., Belgacem et al. 2019; Borhanian & Sathyaprakash 2022; Colombo et al. 2022; Iacovelli et al. 2022; Lehoucq et al. 2023). Interestingly, we find that the population synthesis simulated merger rates deviate by factors up to $20\times$ compared to each of the SFRD models, with the most significant deviations occurring in the range of redshifts $z \gtrsim 3$. These deviations are *significantly* larger compared to those experienced when we scaled the rates by area. Hence, using phenomenological models for the BBH merger rate that scale SFRDs by the local merger rate can lead to the largest deviations from simulated population synthesis merger rates, especially for $z \gtrsim 3$.

3.4. The BHNS and BNS Merger Rate

In Figure 8 and Figure 9, we investigate the merger rate for BHNSs and BNSs, respectively. For both the BHNS and BNS merger rates, deviations from the SFRD are generally similar to those of the BBH merger rate—most of our simulations have a proportionally higher rate of mergers than stars formed at low redshift ($z \lesssim 2$), and a lower rate at high redshift ($z \gtrsim 4$), which are caused by our choice of scaling and the fact that the merger rates peak at lower redshift compared to the SFRD. There are, however, considerably more model-to-model variations for the BNS and BHNS merger rates than the BBH. This stems from the fact that the formation efficiency-metallicity relationship has a markedly greater influence on BHNS and BNS mergers—differing by as much as two to three orders of magnitude—compared to BBH mergers, which vary within an order of magnitude, as can be found in Figure 3. These findings are consistent with similar outcomes reported in the literature (Klencki et al. 2018; Chruslinska et al. 2018; Giacobbo & Mapelli 2018; Neijssel et al. 2019; Román-Garza et al. 2021; Broekgaarden et al. 2022). In our simulations, BHNS and BNS mergers are more commonly formed by binaries that experience a CE phase than BBHs, leading BHNS and BNS simulations to be particularly sensitive to the value of α_{CE} .

Interestingly, we find that many of the BHNS and BNS merger rates in Figure 8 and Figure 9 demonstrate significantly different behavior from the BBH rates and SFRD-based toy models. For example, most of the BHNS merger rates have a significantly steeper drop off after the merger rate peak in Figure 8. In addition, some of the BNS models in Figure 9 show evidence of an additional break in the merger rate slope before the peak, such as the model with $\alpha_{\text{CE}} = 0.1$ and $\beta = 0.25$ around $z = 0.5$. This phenomenon is due to the complex convolution of the SFRD, formation efficiency of BHNSs/BNSs, and delay time distributions, and it lends

weight to considering merger rates that deviate from a SFRD-like rate in future studies (cf. Callister & Farr 2023; Payne & Thrane 2023).

4. DISCUSSION

4.1. Other Formation Channels

This study has investigated the redshift evolution of the merger rate of BBHs (and BNSs and BHNSs) formed by the isolated binary evolution channel. Formation pathways beyond the isolated binary evolution channel, however, can contribute and impact the BBH merger rate as a function of redshift beyond the effects discussed in this paper (e.g., Zevin et al. 2021; Franciolini et al. 2022; Mandel & Broekgaarden 2022; Cheng et al. 2023)⁵. Studying their effects in detail is beyond the scope of this paper, but we discuss the dominant redshift behavior of these formation channels below.

BBHs formed from population III stars (i.e., extremely metal-poor stars) are thought to merger at a significantly different rate than the isolated binary evolution channel: they begin merging at redshifts as early as $z \sim 20$ and peak around $z \sim 10$ (Kinugawa et al. 2020; Belczynski et al. 2017; Hijikawa et al. 2021; Liu & Bromm 2021; Santoliquido et al. 2023). The redshift evolution of population III mergers is still debated and depends strongly on assumed binary formation and interaction mechanisms that can bring black holes close together (e.g. Kinugawa et al. 2014; Kinugawa et al. 2020; Belczynski et al. 2017; Liu & Bromm 2021).

The merger rate for BBHs formed in dense star clusters is also still under debate and is heavily influenced by simulations assumptions and host cluster properties (see Choksi et al. 2019, and references therein). Some studies find that the globular cluster BBH merger rate gradually increases at small redshifts until a peak around $z \sim 2-3$ and a sharp drop-off beyond $z \gtrsim 4$ (see Rodriguez & Loeb 2018 and Choksi et al. 2019). Kritos et al. (2022) and Mapelli et al. (2022) found something similar, but with a longer high-redshift tail, as they assume higher cluster formation rates at high z . Other studies instead find a monotonically increasing function of redshift starting from $z \sim 0$ out to their assumed starting epoch for single burst of globular cluster formation around $z \sim 3-4$, leading to a merger rate that peaks around $z \sim 3-4$ and then drops immediately to zero (Rodriguez et al. 2016; Fragione & Kocsis 2018).

⁵ The evidence for multiple formation channels from observations is still under debate. Recent studies (e.g. Fishbach et al. 2022; Godfrey et al. 2023) find evidence for substantial contributions from the isolated binary evolution channel of BBHs, but large uncertainties remain (e.g. Franciolini et al. 2022; Tong et al. 2022).

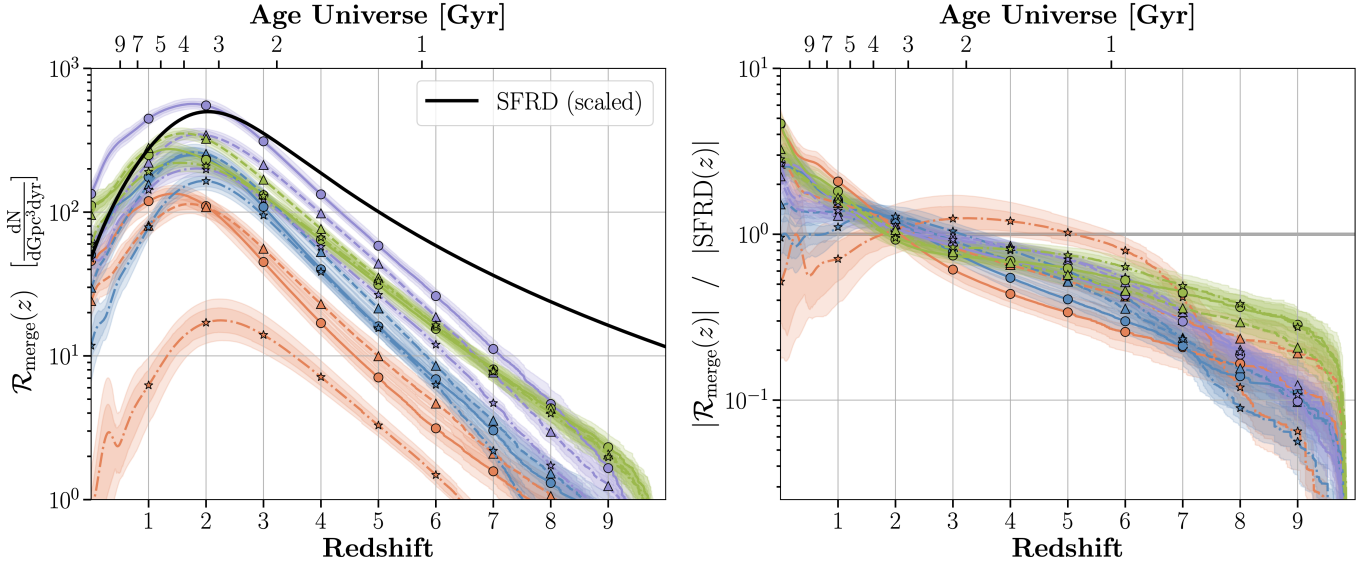


Figure 8. Same as Figure 1 for BHNSs mergers.

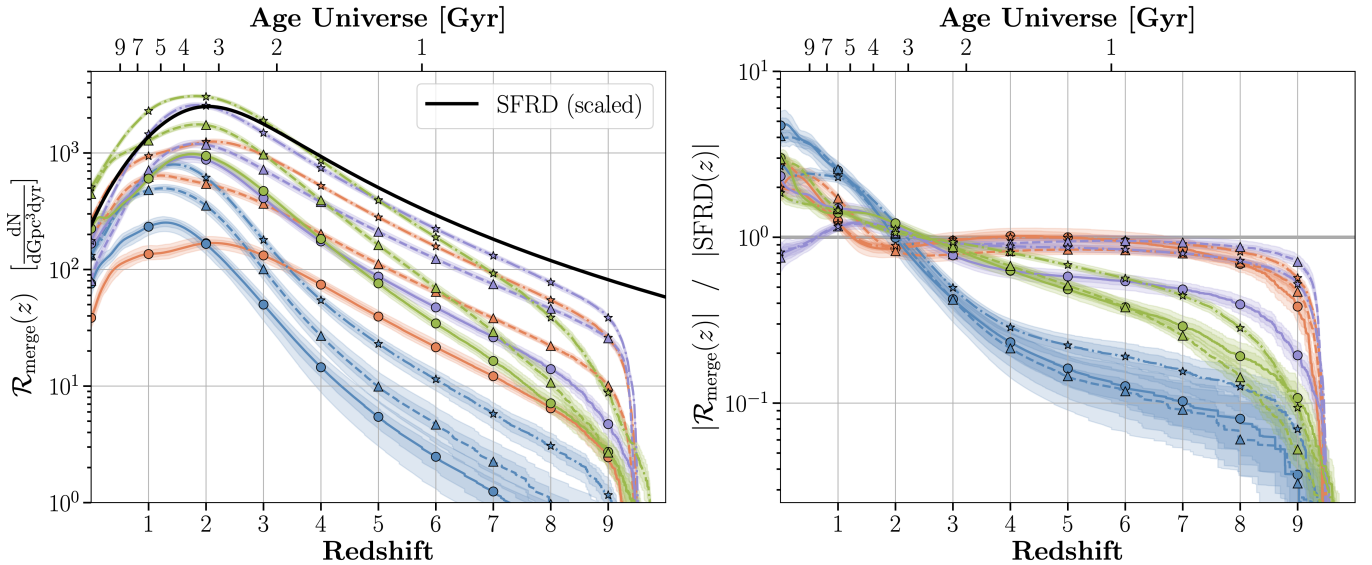


Figure 9. Same as Figure 1 for BNSs mergers.

The sharp decrease in merger rate beyond $z \gtrsim 4$ seen in most simulations is because clusters have not had sufficient time to undergo core-collapse, a process during which BHs sink to the center of clusters and undergo the many dynamical interactions that dominate BBH merger production. Finally, Ye & Fishbach (2024) recently found that the redshift distribution of the BBH merger rate from globular clusters is correlated with primary mass, and that the peak of the merger rate is at higher z for binaries with primary mass $> 30 M_\odot$, and even higher z for primary mass $> 40 M_\odot$.

The merger rate of BBHs formed from chemically homogeneous evolution is still under debate and could ei-

ther lead to a peak earlier or later than isolated binary evolution, mostly depending on the binary interaction and initial conditions assumed in simulation (for more details see de Mink & Mandel 2016; Mandel & de Mink 2016; du Buisson et al. 2020; Riley et al. 2021).

BBHs formed from primordial origins can merge as early as $z \sim 1000$ and are thought to follow a redshift rate that increases monotonically with increasing redshift as $\mathcal{R}_{\text{merge}} \propto t(z)^{-34/37}$ (e.g., Raidal et al. 2019; De Luca et al. 2020, 2021; Franciolini et al. 2022).

It will be important for future studies to further explore differences between the BBH merger rates different

formation pathways (cf. [Martinez et al. 2020](#); [Zevin et al. 2021](#); [Bavera et al. 2022b](#); [Mapelli et al. 2022](#)).

4.2. Observing the Merger Rate

It is challenging to infer the merger rate as a function of redshift from observations. The *local* BHNS, BNS and BBH merger rates have been constrained using GWs, pulsars binary neutron stars (and the lack of black hole-pulsar binaries), as well as short gamma-ray bursts (host galaxies), and r-process enrichment arguments—nevertheless, large uncertainties in these methods remain (see [Mandel & Broekgaarden 2022](#), and references therein). Observations of the rate as a function of redshift remain scarce. The authors of the latest GWTC-3 catalog population inference study found that the rate in the range $0 \lesssim z \lesssim 1.5$ increases proportionally to $(1+z)^\kappa$ with $\kappa = 2.9^{+1.7}_{-1.8}$ ([Abbott et al. 2023](#)). This value of κ is similar to the slope of the SFRD, indicating that the BBH redshift rate follows the SFRD at low redshift (which was also confirmed by [Edelman et al. 2023](#); [Nitz et al. 2023](#)), though large uncertainties remain due to the limited sample of GW observations and lack of models considered (e.g., the analysis from [Abbott et al. 2023](#) assumed that the BBH mass distribution does not change as a function of redshift). Other studies using different models to analyze the GW data found evidence that the slope of the BBH merger rate might deviate from that of the SFRD ([Callister & Farr 2023](#); [Payne & Thrane 2023](#)). We leave direct comparison to observations for future studies, and refer the reader to the discussion in [Ray et al. \(2023\)](#) for more on this topic.

Besides BBHs, the merger rates of BHNS and BNS systems as a function of redshift will also be constrained by future observing runs and next-generation detectors. The merger rate of BNSs (and potentially BHNSs) can also be constrained from electromagnetic observations of gamma-ray bursts (see e.g., [Berger 2014](#); [Fong et al. 2015](#); [Nugent et al. 2022](#); [Zevin et al. 2022](#), and references therein)—although, in practice, this will require a large sample of sGRB with associated host-galaxies as well as a better understanding of sGRB jet physics.

In coming years, new observing runs, including O4 and O5, are poised to increase the number of BBH merger detections to ~ 500 out to $z \sim 2$, which will help constrain the BBH merger rate as a function of redshift [Abbott et al. \(2018\)](#)⁶. Moreover, next-generation GW detectors, such as Cosmic Explorer and Einstein Telescope, are expected to detect stellar-mass black hole mergers out to (and beyond) redshifts $z \gtrsim 10$ and measure the

merger rate with percent level precision (e.g., [Punturo et al. 2010](#); [Sathyaprakash et al. 2012](#); [Reitze et al. 2019](#); [Maggiore et al. 2020](#); [Evans et al. 2021, 2023](#); [Singh et al. 2022](#); [Gupta et al. 2023](#)).

4.3. Caveats

The behavior of the redshift-dependent merger rate of BHNS, BNS, and BBH binaries is impacted by effects beyond those studied in the scope of the paper. We mention the most important ones here.

First, as mentioned in Section 4.1, formation channels beyond the isolated formation channel can impact the merger rate by contributing additional binaries. Each of these channels comes with their own unique uncertainties, increasing the complexity of understanding the overall merger rate behavior (e.g., [Dominik et al. 2013](#); [Zevin et al. 2021](#); [Franciolini et al. 2022](#); [Mandel & Broekgaarden 2022](#); [Arca Sedda et al. 2023](#), and references therein).

Second, not only the merger rate, but also the properties (i.e., masses, black hole spins, mass ratio) of the compact objects are redshift-dependent as a result of both their birth environments depending on redshift (e.g., metallicity), as well as the varying range of delay times can merge by a given z (e.g., [Dominik et al. 2013](#); [Kinugawa et al. 2020](#); [Mapelli et al. 2022](#); [McKernan et al. 2022](#); [van Son et al. 2022](#)). Future work should further investigate the correlation between the merger rate and properties of NSs/BHs in source binaries as a function of redshift to further aid in understanding the underlying physics and formation channels of mergers (e.g. [Qin et al. 2018](#); [Fishbach et al. 2021](#); [Abbott et al. 2021](#); [Belczynski et al. 2022](#); [Bavera et al. 2022a](#); [Biscoveanu et al. 2022](#); [van Son et al. 2022](#); [Callister & Farr 2023](#)).

Third, the underlying metallicity-dependent star formation history is uncertain, especially at higher redshift $z \gtrsim 4$ (see [Chruślińska & Nelemans 2019](#); [Chruślińska 2022](#), and references therein). Although we expect the qualitative results in this study to be the same for different $\mathcal{S}(Z_i, z)$ models, the quantitative results (i.e. with what factor the merger rate deviates from an SFRD’s redshift evolution) will likely depend on the choice of the metallicity-dependent star formation rate assumed. This is because $\mathcal{S}(Z_i, z)$ describes the number of systems formed at a given metallicity as a function of redshift, and will therefore govern when and how much the drop in BBH formation efficiency (Figure 3) will impact the merger rate distribution.

Fourth, there are many uncertainties in the physics underlying single and binary stellar evolution that can impact our study beyond the parameters considered

⁶ See <https://observing.docs.ligo.org/plan/index.html> for the most up-to-date information.

here. A complete study is beyond the scope of this paper (and any single study), so we refer the interested reader to studies including Mapelli (2021); Belczynski et al. (2022); Santoliquido et al. (2021); Broekgaarden et al. (2022); Mandel & Broekgaarden (2022); Spera et al. (2022) and references therein. The most important effects that can impact this study will be uncertainties relating to the formation efficiency behavior (Figure 3) and/or that drastically alter the delay time distributions of BBH, BHNS, and BNS mergers. In particular, models with weak stellar wind loss have found a more gradual decrease in the BBH merger efficiency at high metallicity (e.g. Broekgaarden et al. 2022). For delay time distributions, the angular momentum transport and mass transfer physics (and effect from radius expansion, e.g. Laplace et al. (2020); Romagnolo et al. (2023)) are important uncertainties that impact the separation at which binaries form and thus the merger time. These should be the focus of future work (e.g. Dorozsmai & Toonen 2022; Agrawal et al. 2023).

5. CONCLUSIONS

In this paper we studied how the BBH merger rate expected from isolated binary evolution deviates from the cosmic star formation rate density, focusing on two key effects: metallicity-dependent formation rates and delay times. To achieve this, we conducted a grid of 4×3 simulations using the population synthesis code COMPAS, varying the CE efficiency and mass transfer efficiency parameters. We compared our simulated BBH rates to different SFRD-based models from the literature, and performed similar analysis with BHNS and BNS mergers from our simulations. Below, we summarize our main findings.

1. Simulated BBH merger rates can deviate significantly (factors up to $3.5\text{--}5\times$) from a merger rate model described by a scaled SFRD (Figure 1 and Table 1).
2. These deviations are caused by simulations experiencing (i) more efficient formation of BBHs at low metallicity, leading to a shift of the BBH formation rate peak to $z \sim 2.5$ as opposed to $z \sim 2$ for the SFRD (Figure 2 and Figure 3) and (ii) a broad distribution of delay times that create time gaps from formation to merger, boosting the merger rate at lower redshifts (Figure 4).
3. Many of our simulations display a redshift-dependence in the BBH delay time distribution, favoring shorter delay times at higher redshifts (Figure 5). This is caused by a complex interplay between metallicity and formation channels

causing more orbital shrinking for binaries formed at low metallicity and high redshift. We find that the common envelope efficiency parameter has a strong impact on this redshift evolution.

4. Our simulated BBH merger rates deviate from SFRD models convolved with delay times (Figure 6 and Table 1). This is because *delayed* SFRD models for the BBH rate fail to include the metallicity dependence of BBH formation found in our simulations which causes a decrease in BBH formation at low z .
5. We find even larger deviations (up to factors $\sim 10\times$) when comparing the simulated BBH merger rates to SFRD models scaled to the local merger rate, with the largest deviations at higher redshifts $z \gtrsim 3$ (Figure 7). This means studies that use SFRD-based that are matched to the local GW merger rate might under- or overestimate the underlying BBH merger rate by an order of magnitude at these higher redshifts. The high- z regime will be of particular relevance in coming years when the frontier of GW observation is pushed to the early years of our Universe.
6. Similarly to BBHs, we find that our simulated BHNS and BNS merger rates deviate from a scaled SFRD rate (Figures 8, 9). We do find, however, that the simulated BHNS and BNS merger rates are more impacted by our model's parameter prescriptions than BBHs.
7. Some of the simulated BHNS and BNS merger rates deviate from a simple SFRD-like redshift evolution parameterized by two slopes and a peak, and may actually include breaks before the peak (e.g. our $\alpha_{\text{CE}} = 0.1$ models in Figure 9).

Overall, we find that the simulated BBH, BHNS, and BNS merger rates of the isolated binary evolution channel can significantly deviate from a scaled cosmic star formation rate. This motivates the use of non-SFRD-based merger rate models for future studies and exploration of the merger population.

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Software: This paper made use of simulations from Boesky et al. (2024) which are publicly available at GW Landscape. To create this dataset, we used the COMPAS rapid binary population synthesis code version 2.31.04, which is available for free at <http://github.com/TeamCOMPAS/COMPAS> (Team COMPAS: Riley et al. 2022). The authors used STROOPWAFEL from (Broekgaarden et al. 2019), publicly available at <https://github.com/FloorBroekgaarden/>

STROOPWAFEL. The authors’ primary programming language was PYTHON from the Python Software Foundation available at <http://www.python.org> (van Rossum 1995). In addition, the following Python packages were used: MATPLOTLIB (Hunter 2007), NUMPY (Harris et al. 2020), SCIPY (Virtanen et al. 2020), IPYTHON/JUPYTER (Perez & Granger 2007; Kluyver et al. 2016), ASTROPY (Collaboration et al. 2018) and HDF5 (Collette 2013).

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APPENDIX

A. THE IMPACT OF METALLICITY ON STELLAR EVOLUTION

Metallicity plays a major role in both single and binary star evolution (Neijssel et al. 2019; Broekgaarden et al. 2022; Chruślińska 2022), predominantly as a direct result of its impact on stellar winds (Vink et al. 2001). Winds from massive stars are driven by spectral lines from metals. Wind strips mass from stars, so stars with high initial metallicity lose significantly more mass in their lifetimes than metal-poor stars. This leads stars born with greater proportions of metals to form NSs instead of BHs. The effects of metallicity on *binary* stellar evolution are numerous and even more complex than for single stars. Overall, stronger stellar winds at higher metallicity lead to a decrease in mergers of all types in our simulations due to a range of indirect effects including mass loss associated with stellar winds leading to loss of angular momentum in a binary. This causes greater widening of binaries at high metallicities, increasing inspiral times (Peters 1964) and the number of systems that disrupt, ultimately leading to a decrease in mergers from metal-rich progenitors. More mass loss through stellar winds also leads to systems with smaller stellar envelopes when the binary enters the CE phase. This also leads to less shrinking and fewer systems that can merge in a Hubble time (see Neijssel et al. 2019; Broekgaarden et al. 2022; Chruślińska 2022, and references therein). Another important effect is that metallicity plays a considerable role in determining stellar radial expansion during the star’s evolution with higher metallicities, typically leading to more radial expansion in our simulations (Hurley et al. 2000, 2002; Romagnolo et al. 2023), increasing the probability of unstable mass transfer thus encouraging stellar mergers (cf. Neijssel et al. 2019).

B. METALLICITY-DEPENDENT STAR FORMATION HISTORY

Figure 10 shows the assumed metallicity-dependent SFRD as a function of redshift, $\mathcal{S}(Z_i, z)$. At high redshifts, most stars form with low birth metallicities $Z_i \lesssim Z_\odot/5$. At higher redshift, around $z \gtrsim 3$, most stars instead form from higher metallicity $Z_i \gtrsim Z_\odot/2$. We showed in Figure 2 that the formation yield of BBH mergers in our simulations drastically drops for stars formed with $Z_i \gtrsim 5$. This figure implies that metallicity effects will therefore lead to a drop in the formation of merging BBHs at redshifts $z \lesssim 3$.

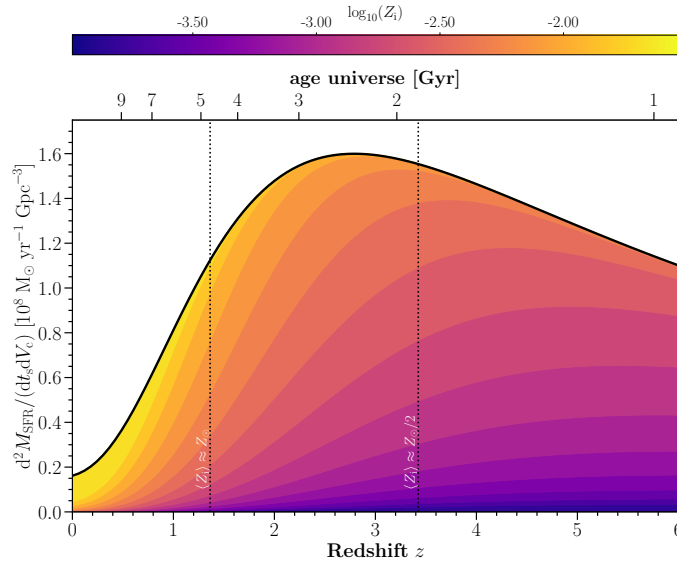


Figure 10. The assumed SFRD colored by stars’ initial metallicity, Z_i . The dotted lines show the location of the median star formation rate for a few metallicities (i.e. half of the stars form with higher metallicity and half with lower metallicity than the labeled value).

C. THE SHAPE OF THE MERGER RATE DUE TO DIFFERENT DELAY TIME MAGNITUDES

In Figure 11, we show how delay times affect a synthetic distribution of stars throughout cosmic history. We rejection sample the SFRD from one million stars out to redshift 10, which we plot in black. We then age the sample stars by 10 Myr, 100 Myr, and 1 Gyr and plot their distributions. In the left panel, we see that the stars travel to lower redshift throughout their lifetimes; in other words, the distributions of the aged stars shift increasingly from the birth distribution toward lower redshift. Furthermore, this effect only causes a notable deviation from the star formation rate when the delay time is of order 500 Myr. A non-trivial aspect of this shift is the fact that the relationship between time and redshift is non-linear. As we can see from the difference between the x-axes on the bottom and top of the panels, time passes faster at higher redshift. This implies that a star born at a high redshift should travel more redshift during its lifetime before merging than if it were born at low redshift.

In the right panel of Figure 11, we take the distributions from the left panel, normalize them, and then divide by the normalized assumed SFRD (similarly to the right panel of Figure 1). The horizontal grey line is what we would observe if the distribution of stars exactly followed the SFRD. Again, we observe that the longer stars are allowed to evolve, the lower redshift they travel to. The stars that were aged by 1 Gyr in particular show the extent to which delay times can cause distributions to shift to lower redshift, as the distribution is far above SFRD (gray line) for redshifts lower than $z \sim 3$, and higher for redshifts above.

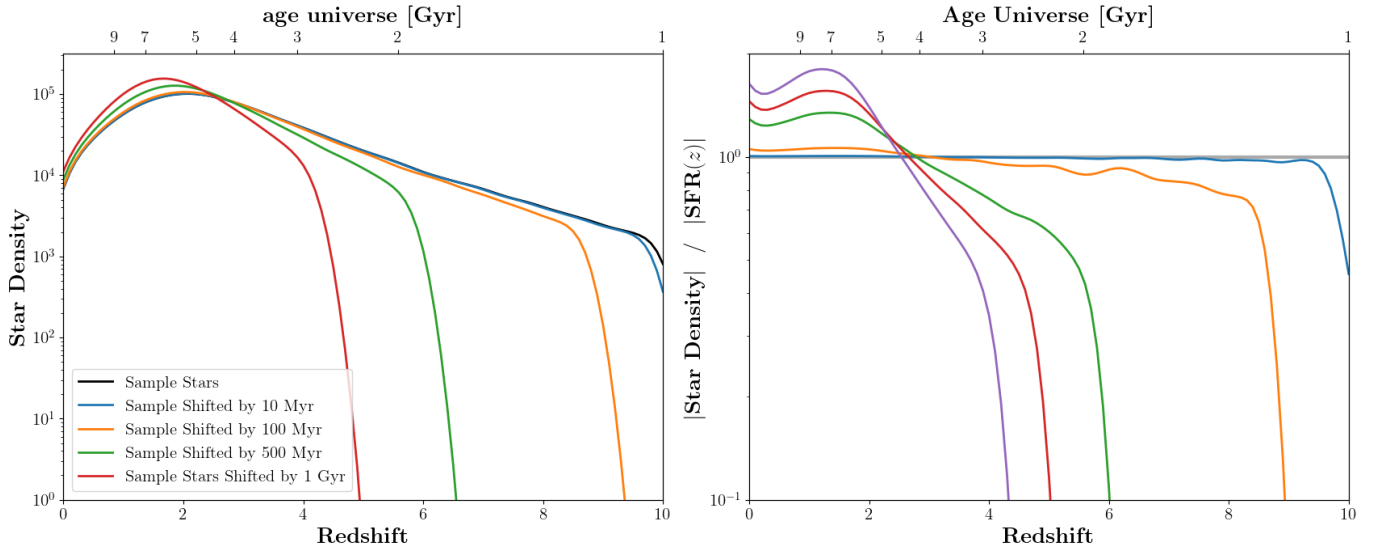


Figure 11. The effect of delay time on the distribution of stars throughout cosmic history. We use rejection sampling of one million stars to acquire a sample of stars reflecting the assumed SFRD (Madau & Fragos 2017). Then, we add 10 Myr, 100 Myr, and 1 Gyr to their birth ages. The left panel shows the original distribution and the distributions of the stars shifted by the three different example delay times. In the right panel, we normalize each distribution and then divide by the normalized SFRD. In doing so, we can see how delay time effects cause the population of stars at low redshift to be greater than the SFRD.

D. BBH FORMATION CHANNELS AS A FUNCTION OF REDSHIFT

As was discussed in the main body of this paper, the delay time distribution is strongly correlated with the formation channels that lead to mergers. In Figure 12, we show the percent contribution of the main formation channels as described in (Broekgaarden et al. 2021) to the merger rate as a function of redshift for all models in grid A. Clearly, the formation pathways that BBH merger progenitors follow are highly dependent on physical assumptions—the contributions of each formation channel to the merger rate are vastly different from model to model. Furthermore, the proportion of mergers created by each channel changes considerably throughout cosmic history, again likely as a result of the difference in the initial metallicity of BBHs progenitors at different redshifts and their interplay with the model prescriptions. Noticeably, trends in channel contribution also appears to be similar for each group of models with the same assumed α_{CE} . This observation is the cause of the delay time distributions being grouped by α_{CE} —the shape of the median delay times throughout cosmic history are similar because the proportion of mergers created by each

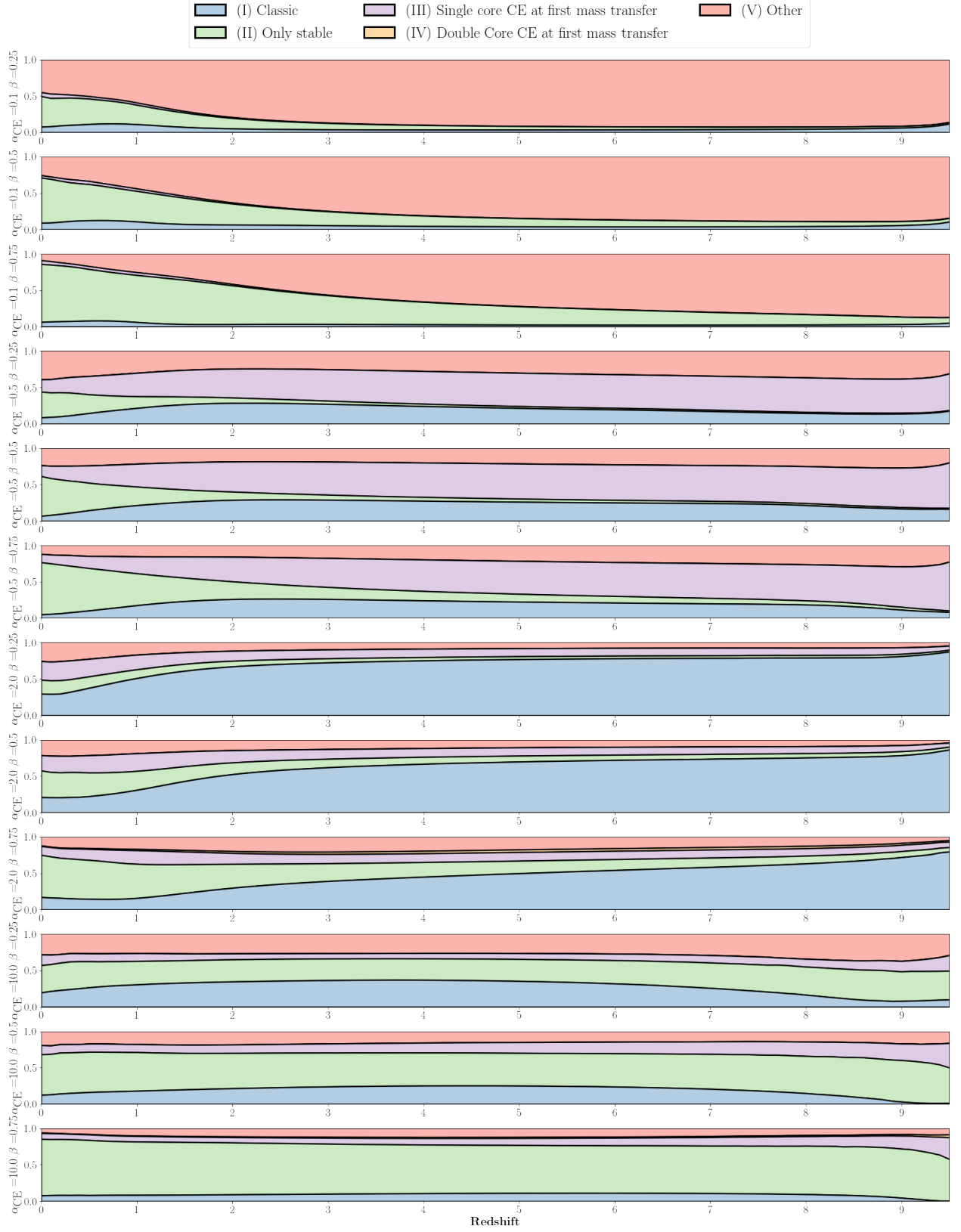


Figure 12. The percent contributions of each formation channel (as described in [Broekgaarden et al. 2021](#)) for the merger rates as a function of redshift of all models in grid A.

Table 1. The minimum, maximum, and median of the normalized BBH merger rates for each model divided by normalized delayed variations of the SFRD for redshift values 0 - 9. The first three columns use a toy model rate by rescaling an SFRD with no delay, the middle three columns use as toy model a SFRD with a constant delay of 20 Myr, and the last three columns use for the toy model a scaled SFRD with a delay time distribution using: t^{-1} with $\min(t_{\text{delay}}) = 20$ Myr.

	No delay			$t_{\text{delay}} \propto \delta(t - 20 \text{ Myr})$			$t_{\text{delay}} \propto 1/t$		
Redshift	Minimum	Median	Maximum	Minimum	Median	Maximum	Minimum	Median	Maximum
$z = 0$	1.27	2.43	3.48	1.74	3.33	4.76	0.21	0.41	0.59
$z = 1$	0.94	1.26	1.62	0.90	1.22	1.56	0.59	0.80	1.02
$z = 2$	0.90	0.95	1.00	0.89	0.94	0.98	1.07	1.13	1.18
$z = 3$	0.79	0.90	1.02	0.78	0.89	1.00	0.97	1.11	1.26
$z = 4$	0.73	0.97	1.10	0.72	0.95	1.09	0.89	1.18	1.35
$z = 5$	0.67	0.99	1.12	0.68	1.01	1.15	0.94	1.39	1.58
$z = 6$	0.59	0.96	1.10	0.63	1.02	1.16	1.03	1.67	1.90
$z = 7$	0.49	0.87	1.01	0.56	0.99	1.16	1.10	1.94	2.26
$z = 8$	0.37	0.70	0.86	0.46	0.88	1.08	1.12	2.16	2.63
$z = 9$	0.20	0.41	0.59	0.30	0.61	0.89	1.03	2.07	3.00

channel is the same. Furthermore, the models with $\alpha_{\text{CE}} = 2.0$, which have the most distinct median delay time shape, have a drastic increase in the proportion of mergers formed through the classic channel (with a common-envelope phase) which corresponds to a drastic decrease in median delay time from $z = 0$ to $z \approx 3$ as at higher redshift the BBH systems form through the classic-CE channel that leads to shorter delay times compared to the only stable mass transfer channel. This is in agreement with studies such as (Olejak et al. 2022; van Son et al. 2022) which found that channels that include CEs tend to have lower delay times. Ultimately, the most important conclusion from this figure is that the delay time distribution is strongly dependent on the channels that the BBH mergers follow, and changes in channel contributions throughout cosmic history are highly non-trivial and dependent on the model assumptions.

E. DEVIATIONS BETWEEN SIMULATIONS AND TOY MODELS

Table 1 shows the deviations at different redshifts between our simulated merger rates and the toy model merger rates.